Satellite Based Emissions Monitoring for the Gulf

Continuous, facility-level air quality tracking powered by Sentinel-5P TROPOMI remote sensing.



PRIMARY SENSOR
Copernicus Sentinel-5P

SPATIAL RESOLUTION
3.5 × 5.5 km

Zero Capex Infrastructure

MARKET CHALLENGE

The GCC Ground Monitoring Gap

Coverage Blind Spots

Sparse surface stations miss vast industrial corridors and offshore platforms.

Detection Lag

Ground sensors report local air quality without identifying plume sources.

Capital Intensity

Physical stations cost tens of thousands per unit, preventing rapid expansion.

Abu Dhabi PM10 Example:

Only 5 primary stations cover the entire metropolitan industrial zone

THE INNOVATION

Project ELARA Solution

1 **Zero-Capex Regional Coverage**

Scans the entire GCC daily using open-access ESA Sentinel-5P data streams.

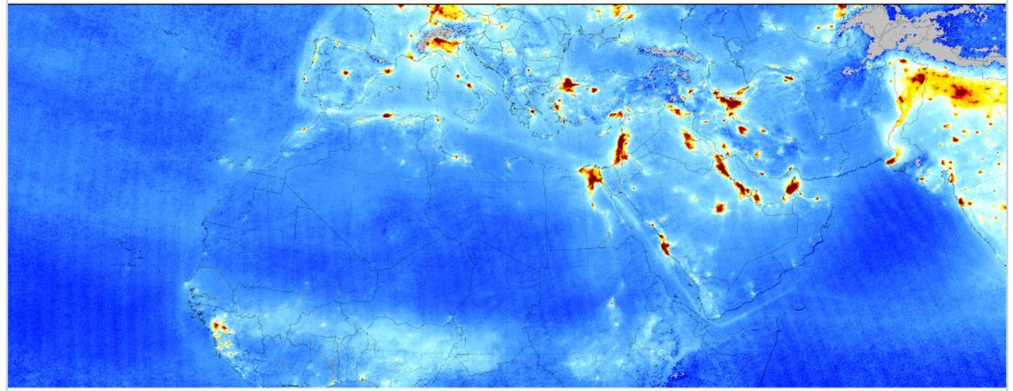
2 **3.5 × 5.5 km Spatial Precision**

Upgraded from legacy OMI resolution to pinpoint specific refineries and ports.

3 **Automated Plume Attribution**

Combines atmospheric gas densities with temporal models for point compliance.

TROPOMI tropospheric NO2, April 2018



TARGET CHEMICAL SPECIES

Atmospheric Gas Portfolio

NO_2

Nitrogen Dioxide

Combustion marker from vehicles, power plants, and oil refining.

SO_2

Sulfur Dioxide

Heavy fuel oil combustion, refining operations, and marine shipping.

HCHO

Formaldehyde

Petrochemical processing marker and ozone precursor.

PM10 / PM2.5

Particulates

Modeled via surface reflectance over unmonitored zones.

Measurement Basis: Tropospheric Column Density integrated vertically (molecules/cm^2 or $\mu mol/m^2$)

OBSERVATIONAL LITERATURE

GCC Pollution Trends (2005–2023)

+12%

Peak Annual NO_2 Surge

Urban and industrial expansion across the Middle East (2005–2014).

−1.7%

Post-2018 Policy Shift

Annual NO_2 decline in Qatar, Bahrain, and Dammam corridor (2018–2023).

+25%

Offshore Platform Spike

Elevation above background levels under calm wind conditions.

Key Takeaway: Stricter vehicular standards and cleaner power shifts caused regional emissions to peak in ~2018, creating a clear policy evaluation baseline.

GOOGLE EARTH ENGINE PIPELINE

Technical Workflow

STEP 01

AOI Filter

Bounding box across Gulf region (40–60°E, 18–32°N).

STEP 02

Data Ingestion

Stream S5P OFFL L3 tropospheric NO_2 collections.

STEP 03

Quality Filter

Threshold qa_value > 0.75 to remove cloud cover distortion.

STEP 04

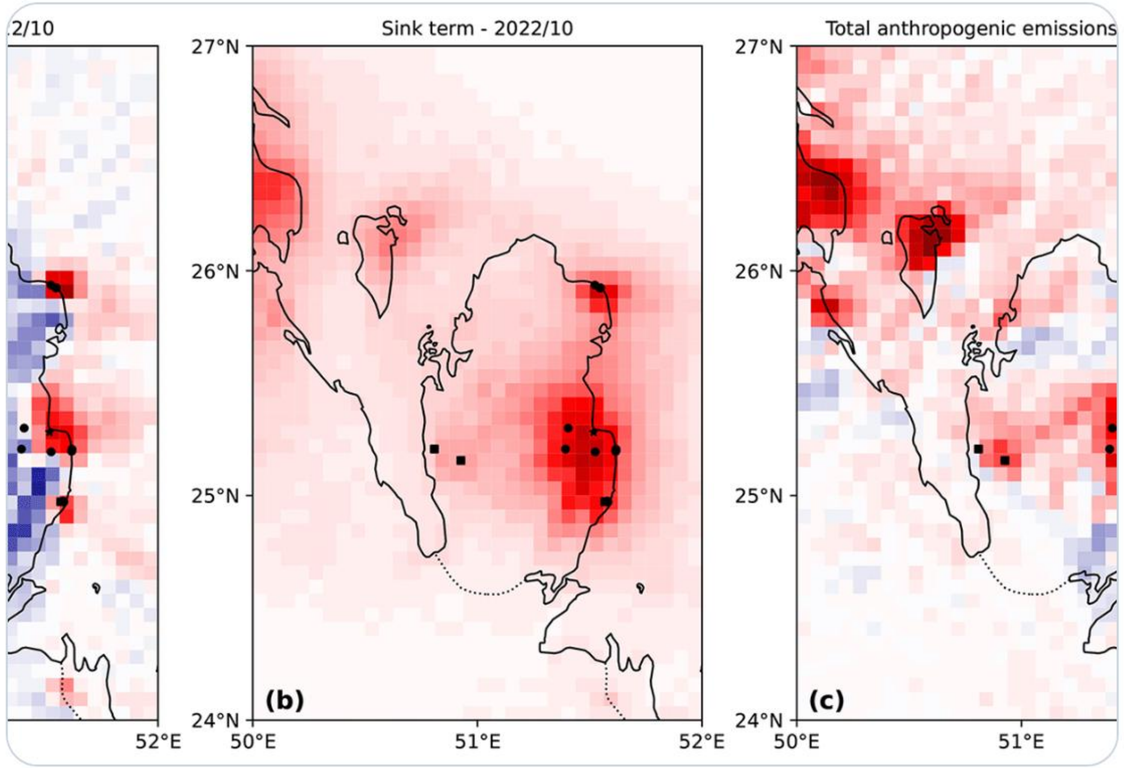
Trend Analysis

Compute multi-year mean composites and percentage deltas.

```
// Earth Engine Sample: var percentageChange = current.subtract(baseline).divide(baseline).multiply(100);
```

PROOF OF CONCEPT OUTPUT

TROPOMI NO_2 Density Mapping



Regional Hotspot Identification

High concentrations highlighted along major refining hubs and shipping lanes.

Point-Source Differentiation

Separates urban traffic plumes from stationary petrochemical stacks.

Model Accuracy Benchmark

Validated within ~5–15% against ship and surface campaigns (SCOAPE).

COMPETITIVE ADVANTAGE

Strategic Value Proposition

Monitoring Approach	Spatial Coverage	Deployment Cost	Attribution Detail
Ground Stations	1–2 km radius around station	$30K–$100K per station	Point specific, but sparse
Manual Compliance Audits	Periodic facility snapshots	High recurring labor	Self-reported, prone to gaps
Generic Air Quality Apps	City-wide regional averages	Low software cost	No facility attribution
Project ELARA Engine	**Full GCC Regional Footprint**	**Near-Zero Hardware Capex**	**Facility Level (3.5 × 5.5 km)**

MARKET ALIGNMENT

Target Customer Ecosystem

Environmental Regulators (B2G)

Regional hotspot dashboards and automated compliance alert systems.

Oil & Gas Operators (B2B)

Self-monitoring reports and early plume leak detection prior to fines.

Industrial Insurers

Historical emissions trend analytics for liability risk pricing.

ESG Consultancies

Independent third-party satellite verification modules.

COMMERCIAL MODEL

Scalable Monetization & Unit Economics

TIER 01 • B2G REGULATORS

Enterprise Dashboards

Annual regional platform licensing with automated compliance alerts and boundary monitoring APIs.

$150K–$300K / yr

Per Government Agency

TIER 02 • B2B OPERATORS

Facility Leak Detection SaaS

Real-time plume tracking and pre-audit compliance reports for refineries and industrial plants.

$25K–$50K / yr

Per Industrial Facility

TIER 03 • ESG & INSURERS

Data-as-a-Service API

Historical emissions risk ratings for underwriting and independent ESG verification auditing.

Usage-Based API

Per Query / Assessment

Target Market (GCC): 15 Regional Agencies + 120 Major Facilities

Software Margins (>85% Gross Margin)

EXECUTION STRATEGY

Technical Roadmap

PHASE 01 • MVP & VALIDATION

Wind Integration

Incorporate ERA5 wind vectors to trace plume origin and cross-validate against surface stations.

PHASE 02 • MULTI-GAS EXPANSION

Methane & SO_2 Modules

Incorporate CH_4 plume detection and integrate hyperspectral data streams.

PHASE 03 • PILOT DEPLOYMENT

Commercial Pilot

Deploy live monitoring dashboard with a regional regulator or oil operator.

”Turning free satellite data into continuous, facility-level emissions records — without installing a single sensor.”

Regulators & Operators
Clear Paying Customers

Fraction of Capex
High ROI vs Physical Sensors

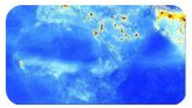
GCC-Wide Scale
Immediate Footprint

Image Sources



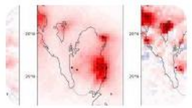
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Source: esoc.esa.int



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Source: www.esa.int



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Source: acp.copernicus.org